

CS1231S TUTORIAL #1

Propositional Logic and Proofs

(AY2025/26 Semester 1) – 11B
Based on Prof Aaron's slides

Introduction



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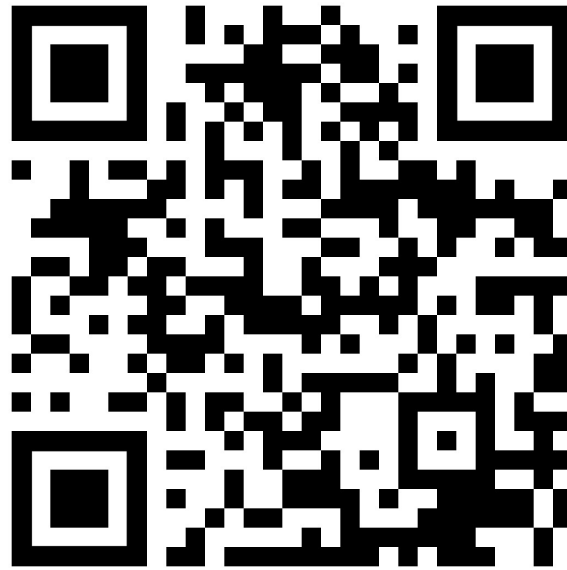
@salmonkarp

Y2 CS

Took CS1231S in 2024/2025 S1
nuSTUDIOS & The Ridge

More introductions!
(while I take attendance)

Telegram Group



<https://t.me/+AZarueRYPVRkMmEg>

Learning objectives for this tutorial:

- Understanding **logical connectives** $\wedge, \vee, \sim, \rightarrow$. (Q1)
- Common mistakes. (Q2)
- Applying **laws of logical equivalences** to simplify statements and to prove equivalence. (Q3, Q4)
- Knowing the **negation**, **contrapositive**, **converse** and **inverse** forms of a conditional statement and their logical relationship. (Q5)
- Definition of the **conditional statement**. (Q6)
- Determining whether an **argument** is valid or invalid. (Q7, Q8)
- Solving knights and knaves problems. (Q9)
- Writing simple **proofs**. (Q10, Q11)

Q1a

$p \equiv$ "I use the umbrella".

$q \equiv$ "It rains".

Propositional statement



I use the umbrella **if** it rains. $q \rightarrow p$

I use the umbrella **only if** it rains. $p \rightarrow q$

Q1b



I use the umbrella **if** it rains.

"I use the umbrella" is a necessary condition for "it rains".

"It rains" is a sufficient condition for "I use the umbrella".

Definition 2.2.7 (Necessary and Sufficient Conditions)

If r and s are statements,

“ r is a sufficient condition for s ” means “if r then s ” or “ $r \rightarrow s$ ”

“ r is a necessary condition for s ” means “if not r then not s ”
or “if s then r ” or “ $s \rightarrow r$ ”

Q1c/d

$p \equiv$ "I use the umbrella".

$q \equiv$ "It rains".

$$(q \rightarrow p) \wedge (p \rightarrow q) \equiv (p \leftrightarrow q)$$



I use the umbrella **if and only if** it rains.

necessary
and
sufficient

"I use the umbrella" is a _____ condition for "it rains".

Q2 (a) $a \wedge \sim(b \wedge c) \equiv a \wedge \sim b \vee \sim c$ by De Morgan's law

is ambiguous!

$a \wedge \sim(b \wedge c) \equiv a \wedge (\sim b \vee \sim c)$ by De Morgan's law

(b) $\sim(x \vee y) \vee z \equiv \sim x \wedge \sim y \vee z$ by De Morgan's law

is ambiguous!

$\sim(x \vee y) \vee z \equiv (\sim x \wedge \sim y) \vee z$ by De Morgan's law

Theorem 2.1.1 Logical Equivalences

For now till midterm test, follow this table and do not skip steps in your answers.

1	Commutative laws	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
2	Associative laws	$p \wedge q \wedge r \equiv (p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$p \vee q \vee r \equiv (p \vee q) \vee r \equiv p \vee (q \vee r)$
3	Distributive laws	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
4	Identity laws	$p \wedge \text{true} \equiv p$	$p \vee \text{false} \equiv p$
5	Negation laws	$p \vee \sim p \equiv \text{true}$	$p \wedge \sim p \equiv \text{false}$
6	Double negative law	$\sim(\sim p) \equiv p$	
7	Idempotent laws	$p \wedge p \equiv p$	$p \vee p \equiv p$
8	Universal bound laws	$p \vee \text{true} \equiv \text{true}$	$p \wedge \text{false} \equiv \text{false}$
9	De Morgan's laws	$\sim(p \wedge q) \equiv \sim p \vee \sim q$	$\sim(p \vee q) \equiv \sim p \wedge \sim q$
10	Absorption laws	$p \vee (p \wedge a) \equiv p$	$p \wedge (p \vee a) \equiv p$
11	Negation of true and false	$\sim \text{true} \equiv \text{false}$	$\sim \text{false} \equiv \text{true}$

Implication law is NOT part of Theorem 2.1.1.

Q4 Prove, or disprove:
 $(p \rightarrow q) \rightarrow r$ is logically equivalent to $p \rightarrow (q \rightarrow r)$.

TRUE
or
FALSE

Definition of conditional
statement $p \rightarrow q$:

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Q4 Prove, or disprove:
 $(p \rightarrow q) \rightarrow r$ is logically equivalent to $p \rightarrow (q \rightarrow r)$.

True or false? If false, what do you need?

What counterexample can you give?



$p \equiv \text{false}; q \equiv \text{true}; r \equiv \text{false}$

$$\begin{aligned}(p \rightarrow q) \rightarrow r \\ &\equiv (F \rightarrow T) \rightarrow F \\ &\equiv T \rightarrow F \equiv F\end{aligned}$$

$$\begin{aligned}p \rightarrow (q \rightarrow r) \\ &\equiv F \rightarrow (T \rightarrow F) \\ &\equiv F \rightarrow F \equiv T\end{aligned}$$

Any of these two
counterexamples
suffices.

$p \equiv q \equiv r \equiv \text{false}$

$$\begin{aligned}(p \rightarrow q) \rightarrow r \\ &\equiv (F \rightarrow F) \rightarrow F \\ &\equiv T \rightarrow F \equiv F\end{aligned}$$

$$\begin{aligned}p \rightarrow (q \rightarrow r) \\ &\equiv F \rightarrow (F \rightarrow F) \\ &\equiv F \rightarrow T \equiv T\end{aligned}$$

Note: Do not follow
these examples for
proof-writing. This is
just a quick sketch.

Q4 Prove, or disprove:
 $(p \rightarrow q) \rightarrow r$ is logically equivalent to $p \rightarrow (q \rightarrow r)$.

If you do not know if it is true or false, you may have to draw the truth table to find out:



p	q	r	$p \rightarrow q$	$(p \rightarrow q) \rightarrow r$	$q \rightarrow r$	$p \rightarrow (q \rightarrow r)$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	T	T	T
T	F	F	F	T	T	T
F	T	T	T	T	T	T
F	T	F	T	F	F	T
F	F	T	T	T	T	T
F	F	F	T	F	T	T

Q3 (a) $\sim a \wedge (\sim a \rightarrow (b \wedge a))$

$$\sim a \wedge (\sim a \rightarrow (b \wedge a))$$

$$\equiv \sim a \wedge (\sim(\sim a) \vee (b \wedge a))$$

$$\equiv \sim a \wedge (a \vee (b \wedge a))$$

$$\equiv \sim a \wedge (a \vee (a \wedge b))$$

$$\equiv \sim a \wedge a$$

$$\equiv a \wedge \sim a$$

$$\equiv \text{false}$$

by the implication law

by the ~~implication law~~

double negative law

by the commutative law

by the absorption law

by the commutative law

by the negation law

Q3 (b) $(p \vee \sim q) \rightarrow q$

$$\begin{aligned} & (p \vee \sim q) \rightarrow q \\ \equiv & \sim(p \vee \sim q) \vee q && \text{by the implication law (step 1)} \\ \equiv & (\sim p \wedge \sim(\sim q)) \vee q && \text{by De Morgan's law (step 2)} \\ \equiv & (\sim p \wedge q) \vee q && \text{by the double negative law (step 3)} \\ \equiv & q \vee (\sim p \wedge q) && \text{by the commutative law (step 4)} \\ \equiv & q \vee (q \wedge \sim p) && \text{by the commutative law (step 5)} \\ \equiv & q && \text{by the absorption law (step 6)} \end{aligned}$$

Q3 (b) $(p \vee \sim q) \rightarrow q$

You may combine two simple steps into one line, by citing the laws applied. Try not to combine more than 2 steps into one line or it may confuse your readers.

$$(p \vee \sim q) \rightarrow q$$

$$\equiv \sim(p \vee \sim q) \vee q \quad \text{by the implication law}$$

$$\equiv (\sim p \wedge \sim(\sim q)) \vee q \quad \text{by De Morgan's law}$$

$$\equiv (\sim p \wedge q) \vee q \quad \text{by the double negative law}$$

~~$$\equiv q \vee (\sim p \wedge q) \quad \text{by the commutative law}$$~~

$$\equiv q \vee (q \wedge \sim p) \quad \text{by the commutative law twice}$$

$$\equiv q \quad \text{by the absorption law}$$

Q3 (c) $\sim(p \vee \sim q) \vee (\sim p \wedge \sim q)$

$$\sim(p \vee \sim q) \vee (\sim p \wedge \sim q)$$

$$\equiv (\sim p \wedge \sim(\sim q)) \vee (\sim p \wedge \sim q) \quad \text{by De Morgan's law}$$

$$\equiv (\sim p \wedge q) \vee (\sim p \wedge \sim q) \quad \text{by the double negative law}$$

$$\equiv \sim p \wedge (q \vee \sim q) \quad \text{by the distributive law}$$

$$\equiv \sim p \wedge \mathbf{true} \quad \text{by the negation law}$$

$$\equiv \sim p \quad \text{by the identity law}$$

Q3 (d) $(p \rightarrow q) \rightarrow r$

$$(p \rightarrow q) \rightarrow r$$

$$\equiv (\sim p \vee q) \rightarrow r$$

by the implication law

$$\equiv (\sim(\sim p \vee q)) \vee r$$

by the implication law

$$\equiv (\sim(\sim p) \wedge \sim q) \vee r$$

by De Morgan's law

$$\equiv (p \wedge \sim q) \vee r$$

by the double negative law

Q5 Given the conditional statement “If $12x - 7 = 29$, then $x = 3$ ”.

Negation: $12x - 7 = 29$ and $x \neq 3$.

Contrapositive: If $x \neq 3$, then $12x - 7 \neq 29$.

Converse: If $x = 3$, then $12x - 7 = 29$.

Inverse: If $12x - 7 \neq 29$, then $x \neq 3$.

Is the given conditional statement true? Yes.

Is the converse true? Yes.

In general, possible for converse to be true but inverse false? No.

Q5 Given the conditional statement “If $12x - 7 = 29$, then $x = 3$ ”.

Proof for “If $12x - 7 = 29$, then $x = 3$ ”:

1. $12x - 7 = 29$ (given)
2. Then $12x = 29 + 7 = 36$ (by basic algebra)
3. Then $x = 3$ (by basic algebra)

Proof for “If $x = 3$, then $12x - 7 = 29$ ”:

1. Substitute $x = 3$ into $12x - 7 = 29$
2. $12(3) - 7 = 36 - 7 = 29$ (by basic algebra)

Note: Converse is logically equivalent to the inverse.

Converse of $p \rightarrow q$: $q \rightarrow p$

Inverse of $p \rightarrow q$: $\sim p \rightarrow \sim q \equiv q \rightarrow p$ (by contraposition)

Q6

$$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$$

p	q	$p \rightarrow_a q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$p \rightarrow_b q$
T	T	T
T	F	F
F	T	T
F	F	F

p	q	$p \rightarrow_c q$
T	T	T
T	F	F
F	T	F
F	F	T

A correct definition of conditional statement $p \rightarrow q$ would result in a tautology for $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$.



So, do the 3 versions above result in $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$ being a tautology if we substitute $\rightarrow$ with $\rightarrow_a, \rightarrow_b, \rightarrow_c$?

Q6

$$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$$

p	q	$p \rightarrow_a q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	r	$p \rightarrow_a q$	$q \rightarrow_a r$	$p \rightarrow_a r$	$(p \rightarrow_a q) \wedge (q \rightarrow_a r)$
T	T	F	T	F	F	F

Conclusion:

$((p \rightarrow_a q) \wedge (q \rightarrow_a r)) \rightarrow_a (p \rightarrow_a r)$
is not a tautology.

$$((p \rightarrow_a q) \wedge (q \rightarrow_a r)) \rightarrow_a (p \rightarrow_a r)$$

F

Q6

$$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$$

p	q	$p \rightarrow_b q$
T	T	T
T	F	F
F	T	T
F	F	F

p	q	r	$p \rightarrow_b q$	$q \rightarrow_b r$	$p \rightarrow_b r$	$(p \rightarrow_b q) \wedge (q \rightarrow_b r)$
T	T	F	T	F	F	F

Conclusion:

$((p \rightarrow_b q) \wedge (q \rightarrow_b r)) \rightarrow_b (p \rightarrow_b r)$
is not a tautology.

$$((p \rightarrow_b q) \wedge (q \rightarrow_b r)) \rightarrow_b (p \rightarrow_b r)$$

F

Q6

$$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$$

p	q	$p \rightarrow_c q$
T	T	T
T	F	F
F	T	F
F	F	T

p	q	r	$p \rightarrow_c q$	$q \rightarrow_c r$	$p \rightarrow_c r$	$(p \rightarrow_c q) \wedge (q \rightarrow_c r)$
F	T	F	F	F	T	F

Conclusion:

$((p \rightarrow_c q) \wedge (q \rightarrow_c r)) \rightarrow_c (p \rightarrow_c r)$
is not a tautology.

$((p \rightarrow_c q) \wedge (q \rightarrow_c r)) \rightarrow_c (p \rightarrow_c r)$
F

Arguments: Valid vs. Sound

To say that an argument form is **valid** means that no matter what particular statements are substituted for the statement variables in its premises, if the resulting premises are all true, then the conclusion is also true.

Definition 2.3.2 (Sound and Unsound Arguments)

An argument is called **sound** if, and only if, it is valid and all its premises are true.

An argument that is not sound is called **unsound**.

Arguments: Valid vs. Sound

Testing an Argument Form for Validity

1. Identify the premises and conclusion of the argument form.
2. Construct a truth table showing the truth values of all the premises and the conclusion.
3. A row of the truth table in which all the premises are true is called a **critical row**.
 - If there is a critical row in which the conclusion is false
 $\Rightarrow$ the argument form is invalid.
 - If the conclusion in every critical row is true
 $\Rightarrow$ the argument form is valid.

Or, deduce through rules of inference.

Rules of Inference

Rule of inference		
Modus Ponens	$p \rightarrow q$ p $\bullet q$	
Modus Tollens	$p \rightarrow q$ $\sim q$ $\bullet \sim p$	
Generalization	p $\bullet p \vee q$	q $\bullet p \vee q$
Specialization	$p \wedge q$ $\bullet p$	$p \wedge q$ $\bullet q$
Conjunction	p q $\bullet p \wedge q$	

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Rule of inference		
Elimination	$p \vee q$ $\sim q$ $\bullet p$	$p \vee q$ $\sim p$ $\bullet q$
Transitivity	$p \rightarrow q$ $q \rightarrow r$ $\bullet p \rightarrow r$	
Proof by Division Into Cases	$p \vee q$ $p \rightarrow r$ $q \rightarrow r$ $\bullet r$	
Contradiction Rule	$\sim p \rightarrow \text{false}$ $\bullet p$	

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Q7 (a) Sandra knows Java and Sandra knows C++.
 $\therefore$ Sandra knows C++.

Let p be "Sandra knows Java".
Let q be "Sandra knows C++".

$p \wedge q$	(premise)
$\therefore q$	(valid by specialization)

Q7

(b) If at least one of these two numbers is divisible by 6, then the product of these two numbers is divisible by 6.
Neither of these two numbers is divisible by 6.
 $\therefore$ The product of these two numbers is not divisible by 6.

Let p be “the first number is divisible by 6”.

Let q be “the second number is divisible by 6”.

Let r be “the product of these two numbers is divisible by 6”.

$p \vee q \rightarrow r$ (premise)

$\sim p \wedge \sim q$ (premise) ($\equiv \sim(p \vee q)$)

$\therefore \sim r$ (invalid: inverse error;
counter-example: 2 and 3.)

Q7

(b) If at least one of these two numbers is divisible by 6, then the product of these two numbers is divisible by 6.
Neither of these two numbers is divisible by 6.
 $\therefore$ The product of these two numbers is not divisible by 6.

Let's discuss this version:

Let x be "at least one of these two numbers is divisible by 6".

Let y be "the product of these two numbers is divisible by 6".

$x \rightarrow y$ (premise)

$\sim x$ (premise)

$\therefore \sim y$ (invalid: inverse error;
counter-example: 2 and 3.)

Which version is better?

Q7

(c) If there are as many rational numbers as there are irrational numbers, then the set of all irrational numbers is infinite.
The set of all irrational numbers is infinite.
 $\therefore$ There are as many rational numbers as there are irrational numbers.

Let p be “there are as many rational numbers as there are irrational numbers”.

Let q be “the set of all irrational numbers is infinite”.

$p \rightarrow q$	(premise)
q	(premise)
$\therefore p$	(invalid: converse error.)

Q7

(d) If I get a Christmas bonus, I'll buy a stereo.
 If I sell my motorcycle, I'll buy a stereo.
 $\therefore$ If I get a Christmas bonus or I sell my motorcycle, I'll buy a stereo.

Let p be "I get a Christmas bonus".
 Let q be "I sell my motorcycle".
 Let r be "I'll buy a stereo".

Many students cite Proof by Division into Cases (Lecture #2 slide 66). This is NOT proof by division into cases.

$p \rightarrow r$	(premise)
$q \rightarrow r$	(premise)
$(p \rightarrow r) \wedge (q \rightarrow r)$	(conjunction premises)
$(\sim p \vee r) \wedge (\sim q \vee r)$	(implication law)
$(\sim p \wedge \sim q) \vee r$	(distributive law)
$\sim(p \vee q) \vee r$	(De Morgan's law)
$\therefore (p \vee q) \rightarrow r$	(implication law)

How do
you fill in
the gap?

Note that for brevity, some steps are skipped. See answer document for the complete answer.

Q8

(a) Given the following argument:

$$p \vee (q \wedge r)$$

$$\sim p$$

$$\therefore q \wedge r$$

Without actually drawing the truth table, determine the values of p , q and r in the critical row(s) of the truth table.

$$p \equiv \text{false}$$

$$q \equiv \text{true}$$

$$r \equiv \text{true}$$

$$q \wedge r \equiv \text{true}$$

Is the argument valid?

Yes

Q8 (b) Give a counterexample to show that the following argument is invalid:

$$p \vee (q \wedge r)$$

$$\sim(p \wedge q)$$

$$\therefore r$$

$$p \equiv \text{true}$$

$$q \equiv \text{false}$$

$$r \equiv \text{false}$$

Q8

(c) Determine whether the following argument is valid or invalid.

If I go to the beach, I will take my shades or my sunscreen.

I am taking my shades but not my sunscreen.

Therefore, I will go to the beach.

Let p be "I go to the beach";

q be "I take my shades";

r be "I take my sunscreen".

$$p \rightarrow q \vee r$$

$$q \wedge \sim r$$

$$\therefore p$$

Critical row: $p \equiv false$; $q \equiv true$; $r \equiv false$.

Conclusion: false.

Argument is **invalid**.

Q8

(c)

$$p \rightarrow q \wedge r$$

$$q \wedge \sim r$$

$$\therefore p$$

p	q	r	$\sim r$	$q \vee r$	$p \rightarrow (q \vee r)$	$q \wedge \sim r$	p
T	T	T	F	T	T	F	T
T	T	F	T	T	T	T	T
T	F	T	F	T	T	F	T
T	F	F	T	F	F	F	T
F	T	T	F	T	T	F	F
F	T	F	T	T	T	T	F
F	F	T	F	T	T	F	F
F	F	F	T	F	T	F	F

Q8

(d) Determine whether the following argument is valid or invalid.

I will buy a new goat or a used Yugo.

If I buy both a new goat and a used Yugo, I will need a loan.

I bought a used Yugo but I don't need a loan.

Therefore, I didn't buy a new goat.

Let p be "I buy a new goat";

q be "I buy a used Yugo";

r be "I need a loan".

$p \vee q$

$(p \wedge q) \rightarrow r$

$q \wedge \sim r$

$\therefore \sim p$

Critical row: $p \equiv \text{false}$; $q \equiv \text{true}$; $r \equiv \text{false}$.

Conclusion: true.

Argument is **valid**.

Q8

(d)

$p \vee q$
 $(p \wedge q) \rightarrow r$
 $q \wedge \sim r$
 $\therefore \sim p$

p	q	r	$\sim r$	$p \wedge q$	$p \vee q$	$(p \wedge q) \rightarrow r$	$q \wedge \sim r$	$\sim p$
T	T	T	F	T	T	T	F	F
T	T	F	T	T	T	F	T	F
T	F	T	F	F	T	T	F	F
T	F	F	T	F	T	T	F	F
F	T	T	F	F	T	T	F	T
F	T	F	T	F	T	T	T	T
F	F	T	F	F	F	T	F	T
F	F	F	T	F	F	T	F	T

Q10 Prove:
The product of any two odd integers is an odd integer.

Direct proof:

Give justification

1. Take any two odd integers n, m .
2. Then $n = 2k + 1$ and $m = 2p + 1$, for $k, p \in \mathbb{Z}$ (by definition of odd integers)
3. Hence $nm = (2k + 1)(2p + 1)$
$$= (2k(2p + 1)) + (2p + 1)$$
$$= (4kp + 2k) + (2p + 1)$$
$$= 2(2kp + k + p) + 1 \text{ (by basic algebra)}$$
4. Let $q = 2kp + k + p$ which is an integer (by closure of integers under $+$ and $\times$).
5. Then $nm = 2q + 1$ which is odd (by definition of odd integers)
6. Therefore, the product of any two odd integers is an odd integer.

Q11 Let n be an integer. Then n^2 is odd if and only if n is odd.

(a) Smart's attempt:

Proof (by contradiction) ?

Missing justifications.

1. Suppose n is an even integer.
2. Then $\exists k \in \mathbb{Z}$ s.t. $n = 2k$. (by definition of even integers)
3. Squaring both sides, we get $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$.
4. Since k is an integer, so is $2k^2$. (by closure of integers under $\times$)
5. Hence $n^2 = 2p$, with $p = 2k^2 \in \mathbb{Z}$.
6. Therefore, n^2 is even. (by definition of even integers)
7. So, if n is even, then n^2 is even, which is the same as saying, if n^2 is odd, then n is odd.
8. Therefore, n^2 is odd if and only if n is odd.

Prove only one direction.

Q11 Let n be an integer. Then n^2 is odd if and only if n is odd.
(b) Write your own proof.

Proof:

1. ($\Rightarrow$) Proving the contraposition of “if n^2 is odd, then n is odd”.
 - 1.1. Suppose n is even.
 - 1.2. Then $\exists k \in \mathbb{Z}$ s.t. $n = 2k$. (by definition of even integers)
 - 1.3. Squaring both sides, we get $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$. (by basic algebra)
 - 1.4. Hence $n^2 = 2p$, with $p = 2k^2 \in \mathbb{Z}$. (by closure of integers under $\times$)
 - 1.5. Therefore, n^2 is even. (by definition of even integers)
 - 1.6. This proves that if n^2 is odd, then n is odd.
2. ($\Leftarrow$) If n is odd, then $n \times n = n^2$ is odd. (by question 10)
3. Therefore, n^2 is odd if and only if n is odd. (by lines 1.6 and 2)

Q9

- (a) Two natives A and B :
 A says: Both of us are knights.
 B says: A is a knave.



Proof (by contradiction):

1. If A is a knight, then:
 - 1.1 What A says is true. (by definition of knight)
 - 1.2 $\therefore B$ is a knight too. (that's what A says)
 - 1.3 $\therefore$ What B says is true. (by definition of knight)
 - 1.4 $\therefore A$ is a knave. (that's what B says)
 - 1.5 $\therefore A$ is not a knight. (since A is either a knight or a knave, but not both)
 - 1.6 $\therefore$ Contradiction to 1.
2. $\therefore A$ is not a knight.
3. $\therefore A$ is a knave. (since A is either a knight or a knave, but not both)
4. $\therefore$ What B says is true.
5. $\therefore B$ cannot be a knave. (as B has said something true)
6. $\therefore B$ is a knight. (as there are only knights and knaves)
7. Conclusion: **A is a knave and B is a knight.**

Tempting to say "Contradiction" right after 1.4. However, this is not valid because contradiction requires $p \wedge \sim p$, but 'knave' is not the negation of 'knight'. Hence 1.5 is required before we arrive at the contradiction in 1.6.

Q9

- (b) Two natives C and D :
 C says: D is a knave.
 D says: C is a knave.
How many knights and knaves are there?



Proof (by division in cases):

1. If C is a knight, then:
 - 1.1 What C says is true. (by definition of knight)
 - 1.2 $\therefore D$ is a knave. (that's what C says)
2. If C is not a knight, then:
 - 2.1 Then C is a knave. (one is either a knight or a knave)
 - 2.2 $\therefore$ what C says is false. (by definition of knave)
 - 2.3 $\therefore D$ is not a knave. (C says D is a knave, but what C says is false)
 - 2.4 $\therefore D$ is a knight. (one is either a knight or a knave)
3. Conclusion: in both cases, there is **one knight and one knave**.

This is actually a rather 'incomplete' proof. While the conclusions are correct, we never check on whether each configuration is consistent or not.

In general, don't forget to check again whether what C/D says is possible *after* you've deduced their identities.

This is actually a rather 'incomplete' proof. While the conclusions are correct, we never check on whether each configuration is consistent or not.

In general, don't forget to **check again whether what C/D says is possible *after* you've deduced their identities.**

For more details, try to solve this instead to see why checking consistency is important.

C: "D is a knight."

D: "Exactly one of us is a knight."

Prove that there are 0 knights.

Tutorial answers will be uploaded on Canvas at the end of the week.

If you still have any doubts, please post on QnA.

